25ECPCC301IAE-II 01 OCT 2026 Student Name : Harsh Pandya

St. John College of Engineering and Management, Palghar, Maharashtra

Autonomous Institute

Internal Assessment Examinations, October 2026

B.Tech. Electronics & Computer Science - SEMESTER III

25ECPCC301 - Mathematics for ECS

Time : 1 Hour **Maximum Marks : 20**

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Instructions

- All Questions are compulsory and carry equal weightage of 5 marks
- Assume suitable data wherever required

Q1	Attempt any 5 out of 6 Questions	Max. marks(5)	Course Outcome	RBT Level
	a)What is a **random variable**?	1	CO4	L1
	b)Write the probability function of a **Poisson distribution**.	1	CO4	L1
	c)What is the formula for the Chi square test Stistics ?	1	CO5	L1
	d)What is Null Hypothesis ?	1	CO5	L1
	e)What is the purpose of Dijkstra's algorithm?	1	CO6	L1
	f)What is an odd function?	1	CO6	L1

Q2 a) Find k if the following function is a p.d.f. $f(x) = \begin{cases} \frac{x}{6} + k, & 0 \le x \le 3, \\ 0, & \text{otherwise.} \end{cases}$ — 5 | CO4 | L3

Find $P(1 \le X \le 2)$.

OR

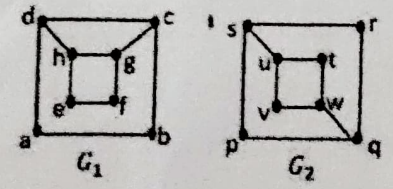
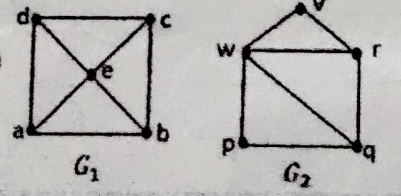
b) A discrete random variable has the probability mass function given below. Find k, the mean and variance. — 5 | CO4 | L3

X	-2	-2	0	1	2	3
$P(X=x)$	0.2	k	0.1	$2k$	0.1	$2k$

Q3 a) — 5 | CO5 | L3

A hospital is testing two antibiotics to treat a particular disease. Medical researchers want to statistically analyze whether there is a significance in patient recovery periods between the two.

	Type A	Type B
No. of Patients	6	6
Mean period (in days)	13.55	10.10
Unbiased standard deviation (in days)	3.2	2.8

	Two independent groups of patients were treated using Antibiotic A and Antibiotic B. The length of recovery (in days) was recorded as above. Using a 5% level of significance, test the Null Hypothesis that the difference in mean recovery periods between the two antibiotics is significant.			
	OR			
	b)In a town there are **2000 literate persons**. Out of them, **60% read newspaper A**, **55% read newspaper B**, and **20% read neither A nor B**. How many persons read: 1. **Both newspaper A and B** 2. **Only one newspaper**	5	CO5	L3
Q4	a)Differentiate between Eulerian and Hamiltonian graphs with examples.	5	CO6	L3
	OR			
	i)  ii)  b) Determine whether the graphs GI and G2 are isomorphic or not. Justify your answer.	5	CO6	L3

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